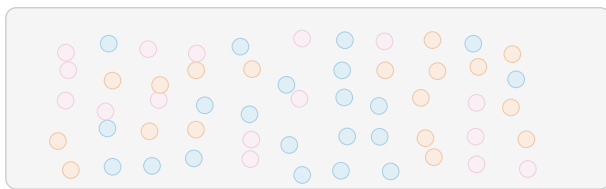


a NOTEARS: Pooled optimization — all samples mixed

Gradient signal diluted into heterogeneous noise

a



$$\min_W \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda_1 \|W\|_1 \quad \text{s.t. } h(W) = 0$$

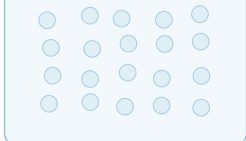
$\Rightarrow$ **High- d failure:** gradient dominated by noise
L1 regularization drives $W \rightarrow \mathbf{0}$, zero edges recovered

b CAGate: Cluster-aware gating — structure-rich subpopulations weighted

MAD-aware gate γ_k amplifies structure-rich clusters, suppresses noise-dominated ones

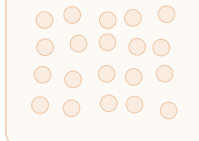
b

Cluster A
High MAD, structure-rich



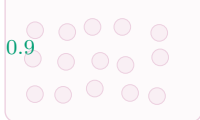
Cluster B

Low MAD, noisy



Cluster C

Intermediate MAD



$$\gamma_A \approx 0.9$$

$$\gamma_B \approx 0.1$$

$$\gamma_C \approx 0.2$$

$$\min_W \frac{1}{2n} \sum_{k=1}^K \gamma_k \cdot \|\mathbf{X}^{(k)} - \mathbf{X}^{(k)}W\|_F^2 + \lambda_1 \|W\|_1$$

$$\text{s.t. } h(W) = 0, \quad \gamma_k = 1 - e^{-\alpha \cdot \overline{\text{MAD}}_k}$$

$\Rightarrow$ **High- d survival:** structure-rich cluster gradients dominate
CAGate recovers 550+ edges where NOTEARS finds zero